Experiment 3: Implementation and analysis of Spatial Filtering Techniques using low-pass and High-pass Filters

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Course & Branch- B.TECH CSE

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SUBJECT - Computer Vision Lab

SEM & Section - 5th SEM , "B"

GitHub Repo Link ->
https://github.com/cu24250181-Ag/Computer_vision_LAB/tree/main/Experiment%203

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files
from google.colab.patches import cv2_imshow

uploaded = files.upload()
image_path = list(uploaded.keys())[0]

img_color = cv2.imread(image_path)
img_color = cv2.cvtColor(img_color, cv2.COLOR_BGR2RGB)
img_gray = cv2.cvtColor(img_color, cv2.COLOR_RGB2GRAY)
```

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```
plt.figure(figsize=(6, 6))
plt.imshow(img_gray, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis('off')
plt.show()
```




```
gaussian_blur = cv2.GaussianBlur(img_gray, (5, 5), 0)

plt.imshow(gaussian_blur, cmap='gray')
plt.title("Gaussian Blur")
plt.axis('off')
plt.show()
```




```
median_blur = cv2.medianBlur(img_gray, 5)

plt.imshow(median_blur, cmap='gray')
plt.title("Median Filter")
plt.axis('off')
plt.show()
```



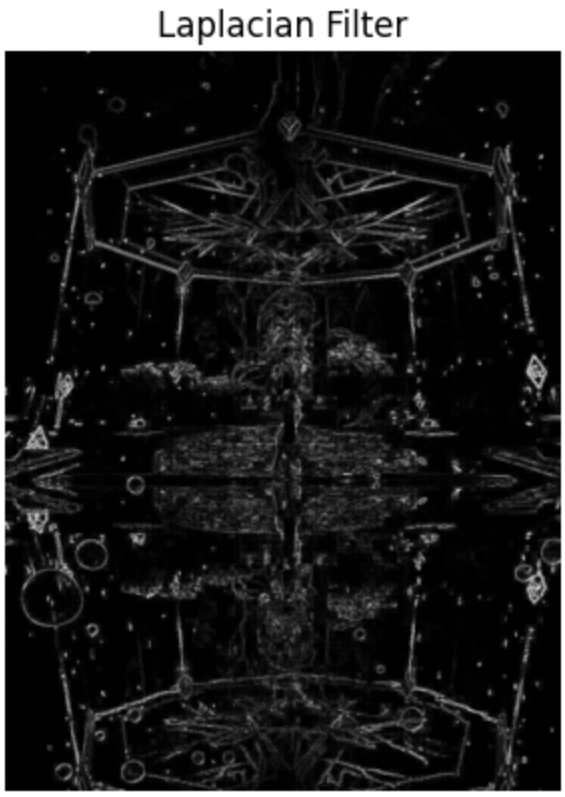

```
average_blur = cv2.blur(img_gray, (5, 5))

plt.imshow(average_blur, cmap='gray')
plt.title("Average Filter")
plt.axis('off')
plt.show()
```



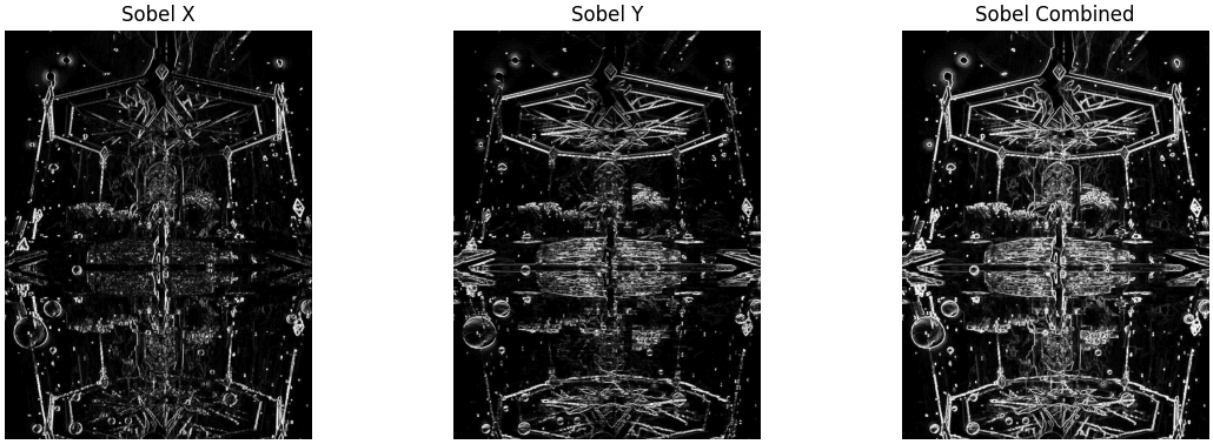

```
laplacian = cv2.Laplacian(img_gray, cv2.CV_64F)
laplacian = cv2.convertScaleAbs(laplacian)

plt.imshow(laplacian, cmap='gray')
plt.title("Laplacian Filter")
plt.axis('off')
plt.show()
```



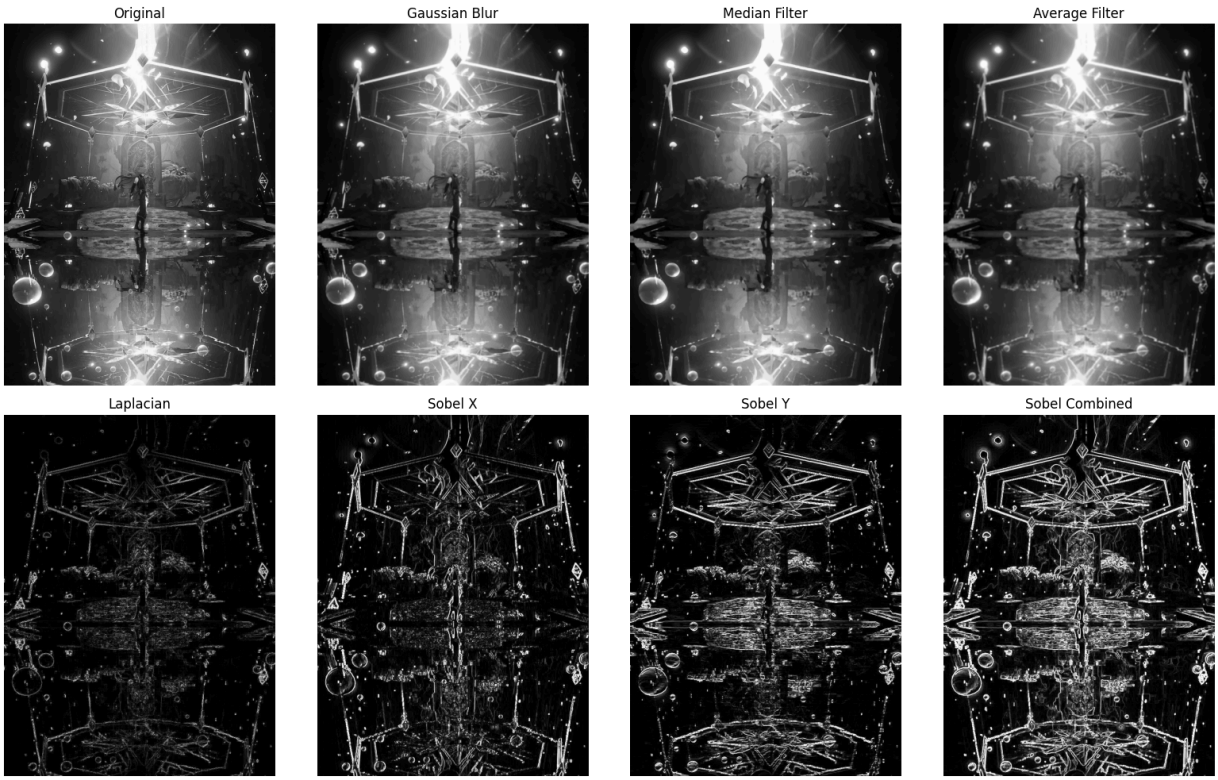

```
sobel_x = cv2.Sobel(img_gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(img_gray, cv2.CV_64F, 0, 1, ksize=3)
sobel_x = cv2.convertScaleAbs(sobel_x)
sobel_y = cv2.convertScaleAbs(sobel_y)
sobel_combined = cv2.bitwise_or(sobel_x, sobel_y)

plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1); plt.imshow(sobel_x, cmap='gray'); plt.title("Sobel X"); p
plt.subplot(1, 3, 2); plt.imshow(sobel_y, cmap='gray'); plt.title("Sobel Y"); p
plt.subplot(1, 3, 3); plt.imshow(sobel_combined, cmap='gray'); plt.title("Sobel
plt.tight_layout()
plt.show()
```




```
titles = ['Original', 'Gaussian Blur', 'Median Filter', 'Average Filter',
          'Laplacian', 'Sobel X', 'Sobel Y', 'Sobel Combined']
images = [img_gray, gaussian_blur, median_blur, average_blur,
          laplacian, sobel_x, sobel_y, sobel_combined]

plt.figure(figsize=(16, 10))
for i in range(len(images)):
    plt.subplot(2, 4, i + 1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')
plt.tight_layout()
plt.show()
```



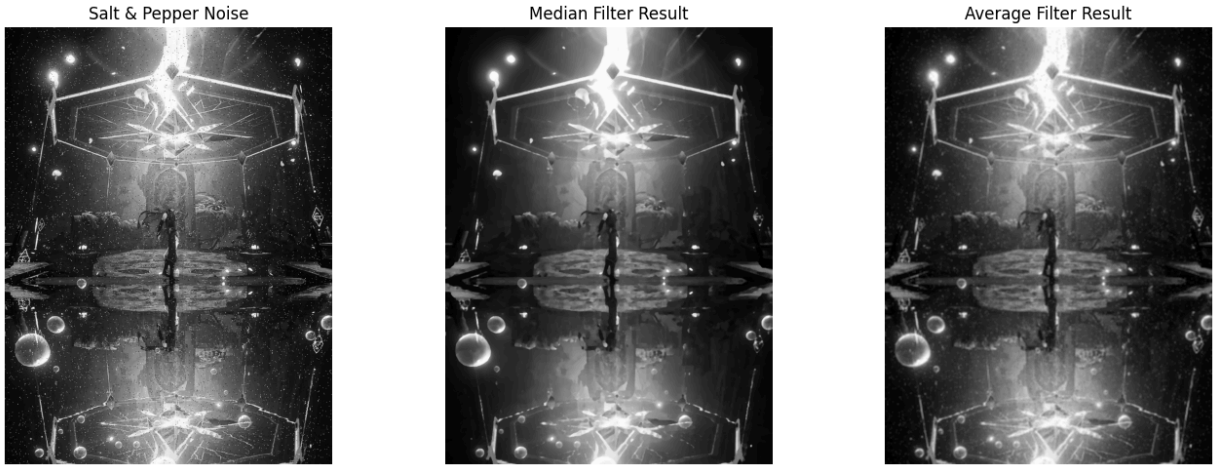

```
def add_salt_pepper_noise(image, amount=0.02):
    noisy = np.copy(image)
    num_salt = np.ceil(amount * image.size * 0.5)
    num_pepper = np.ceil(amount * image.size * 0.5)
```

```
    coords = [np.random.randint(0, i - 1, int(num_salt)) for i in image.shape]
    noisy[coords[0], coords[1]] = 255

    coords = [np.random.randint(0, i - 1, int(num_pepper)) for i in image.shape
    noisy[coords[0], coords[1]] = 0
    return noisy

noisy_img = add_salt_pepper_noise(img_gray)
median_denoised = cv2.medianBlur(noisy_img, 5)
average_denoised = cv2.blur(noisy_img, (5, 5))

plt.figure(figsize=(14, 5))
plt.subplot(1, 3, 1); plt.imshow(noisy_img, cmap='gray'); plt.title("Salt & Pep
plt.subplot(1, 3, 2); plt.imshow(median_denoised, cmap='gray'); plt.title("Medi
plt.subplot(1, 3, 3); plt.imshow(average_denoised, cmap='gray'); plt.title("Ave
plt.tight_layout()
plt.show()
```



Question Answers

1. What is spatial filtering?

Ans-> Spatial filtering is an image enhancement technique that modifies a pixel's value based on the values of its neighboring pixels, using a small matrix called a kernel (or mask) that slides over the image via convolution. It's used for smoothing, sharpening, noise removal, and edge detection directly in the pixel domain (as opposed to transforming to frequency domain first).

2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.

Ans->

- Low-pass filters allow slow variations (smooth regions) to pass through while suppressing high-frequency components like noise and edges — result: blurred/smoothed image. Examples: Gaussian, Average/Mean, Median.
- High-pass filters suppress slow variations and emphasize high-frequency components like edges and fine details — result: sharpened image with enhanced boundaries. Examples: Laplacian, Sobel, Prewitt.

3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications